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Current status of greenhouse gas emissions from aquaculture in China

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ABSTRACT

Aquaculture and mariculture are becoming an increasingly important source of food supply food in many countries and regions. However, with the expansion of aquaculture and mariculture comes increasing emissions of greenhouse gases (GHG) which contribute to global warming and climate change. China leads the world in aquaculture and mariculture production, but there are no studies that systematically assess China's overall carbon footprint from these industries. This study quantified GHG emissions from aquaculture and mariculture by four source phases (feed, energy use, nitrous oxide and fertilizers), and then analyzed the carbon footprint of each of these phases for GHG production of nine major species groups over the past ten years to show the spatial distribution of GHG emissions from aquaculture and mariculture in China. Our results showed that the production of feed materials contributed most to the GHG emissions and found that crop energy use, crop land use changes (LUC), fertilizer production, crop nitrous oxide production and rice methane production were the main sources of feed emissions. The total GHG emissions of the nine species groups were 112 Mt (10^9 kg) CO_2e, the nine species accounting for approximately 86% of aquaculture and mariculture production. GHG emissions of cyprinids had the highest contribution at 47%. Spatial analysis based on our study showed Guangdong, Hubei, Jiangsu and Shandong had the highest GHG emissions of all the provinces in this study, and they accounted for approximately 46% of all emissions. The regional Gross Domestic Product (GDP) was significantly positively correlated with GHG emissions in every province, with a correlation coefficient higher than 0.6. Our results showed for the first time the relationship between the relative production by species composition and spatial distribution of GHG emissions from aquaculture and mariculture in China. Our findings provide the scientific basis for reduction of GHG emissions within a broader context of expanding aquaculture in the future.

1. Introduction

Globally, aquaculture and mariculture are increasingly important as a key agriculture sectors that can make an important contribution to global food security, and to help alleviate poverty in impoverished and poorer regions (MacLeod et al., 2020; FAO, 2018). Aquaculture (freshwater fishery) and mariculture (marine fishery) as an alternative source to traditional food production systems, can provide more food for an increasing global human population (Cao et al., 2011). Global production from aquaculture and mariculture accounts for 50% of the world's fish supply (FAO, 2010; Cao et al., 2013). In recent decades, these industries have become the most rapidly increasing food production sector (Bostock

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et al., 2010) and still has great potential to grow at a significant rate (Diana, 2009). However, with the expansion of aquaculture and mariculture, it has been suggested that this will lead to more environmental problems, such as eutrophication, water pollution, threats to biodiversity, land use changes (LUC), and increased greenhouse gas (GHG) emissions (Mazzola et al., 2000; Read and Fernandes, 2003; Ray et al., 2019).

GHG emissions are one of the most important drivers of global climate change. GHGs include nitrous oxide (N_2O), carbon dioxide (CO_2) and methane (CH_4), production of which are all components of aquaculture and mariculture production (MacLeod et al., 2019). Nitrous oxide comes mainly from microbial transformation of nitrogen in soils during cultivation of crops but also from microbial transformation of nitrogenous compounds derived from feed and fertilizer in aquaculture ponds (Kosten et al., 2020; MacLeod et al., 2020). Methane arises mainly from anaerobic decomposition of organic material during rice cultivation. The emissions of CO_2 from aquaculture is very wide, ranging from the construction of pre-farms to the use of energy and feeding bait on-farms to the transportation and sales of aquatic products post-farms (MacLeod et al., 2020). Anthropogenic GHG emissions from activities such as aquaculture, lead to serious consequences such as climate change and ocean acidification (Doney et al., 2012). Therefore, taking immediate action to mitigate GHG emissions is a priority for the aquaculture and mariculture industries around the world.

China is the world's leading producer of food from aquaculture and mariculture, accounting for 48% and 62% of the world's production in 2008 and 2016 respectively (Bostock et al., 2010; FAO, 2016; Ren et al., 2019). China's aquaculture and mariculture industries are diverse, and spread throughout the country. In inland provinces, the main farmed species are tilapia (*Oreochromis* spp.), freshwater prawn (mainly *Macrobrachium rosenbergii*), Chinese perch (*Siniperca chuatsi*), soft-shelled turtle (*Pelodiscus sinensis*) and the river eel (mainly *Anguilla japonica*) (Zhao et al., 2019). The common sea perch (*Lateolabrax japonicus*), the large yellow croaker (*Larimichthys crocea*) and Whiteshrimp (*Litopenaeus Vannamei*) are the main mariculture species farmed in coastal provinces. The proportion of food in diets derived from aquaculture and mariculture is gradually increasing, but there are only a few analyses of the carbon footprint of these industries in China. For instance, Jin and Chen, 2012 assessed the carbon emissions of aquaculture from on-farm energy use and compared the carbon emissions from different provinces. Chang et al. (2017) used life cycle assessment (LCA) to analyze the carbon footprint of an ecological shrimp farm in Taiwan and Liu et al. (2019) systematically reviewed existing research on GHG emmisions from freshwater ponds in China and abroad. Despite China being the largest producer of food from aquaculture and mariculture. To date, there are no studies to systematically assess China's overall carbon footprint from aquaculture or mariculture. We have quantified the contribution to China's aquaculture and mariculture carbon footprint for each of the most commonly cultured species including catfish, cyprinids, shrimp, tilapia, bivalves, marine and other freshwater fish, and analyzed how their impacts can be mitigated. We also calculated the emission intensity (EI, Mt CO_2e/Mt production) of aquaculture and mariculture, and identified and assessed the factors that influence it.

This study quantified the carbon footprint from China's average annual aquaculture and mariculture production from 2011 to 2020, and focuses on the GHG emissions from four production phases—feed raw materials; on-farm energy use; aquatic N_2O production; and fertilizers. The objectives of this study were to (1) quantify the average GHG emissions, and EI from aquaculture and mariculture over the past 10 years in China, (2) to analyze the composition of the GHG emissions and EI from the nine target species groups, (3) to analyze the spatial distribution of the GHG emissions at the province scale in China, and (4) to understand the nature of the relationship between GHG emissions and regional Gross Domestic Product (GDP). The analysis carried out in this study contributes to our understanding of GHG emissions emanating from aquaculture and mariculture in China. It provides the evidence-base for the interventions required to reduce GHG emissions from aquaculture and mariculture now, and as these industries expand into the future.

2. Materials and methods

2.1. Scope

The average carbon footprint was calculated for 2011 to 2020 and following four phases of production: (1) GHG emissions from feed, including the emissions from production of raw feed materials, processing and transport of feed materials, and production of compound feed in feed mills and transport to the fish farm; (2) GHG emissions for on-farm energy use; (3) GHG emissions for aquatic N_2O production; and (4) GHG emissions from fertilizers. (MacLeod et al., 2020). The species groups used by MacLeod et al. (2020) were used to select categories of bivalve, catfish, cyprinids, general freshwater fish, marine fish, shrimp and prawns, and tilapia production extracted from the China Fisheries Statistical Yearbooks (2012–2021) and broken down to province and species levels.

2.2. GHG emissions

2.2.1. GHG emissions for feed

Based on the feed conversion ratios as shown by MacLeod et al. (2020), the total weight of feed consumed was calculated for each species using Equation (1) thus:

$$F_i = FCR_i * P_i \quad (1)$$

where F_i is the total weight (kg) of the feed for each species i, FCR_i is the feed conversion ratio of species i, and P_i is the production (kg) of species i. Because aquafeed is comprised of both commercial and farm-made/semi-commercial feeds with different emission factors, the proportion of total weight of commercial and non-commercial aquafeed was estimated based on data from Greenpeace (2017). Because the composition (such as fishmeal, wheat and soybean meal) of commercial and non-commercial aquafeed is different in different cities and regions, the weight of raw material of each feed was also calculated according to MacLeod et al. (2020), and the GHG emissions of each feed raw material calculated using Equation (2) thus:

$$GE_{fs} = EF_{fs} \times AC_{fs} \quad (2)$$

where GE_{fs} is the mass (kg) of GHG emissions from feed raw material s, EF_{fs} is the emission factor of feed raw material s (MacLeod et al., 2020), AC_{fs} is the amount of each raw feed material, and all the GHG emissions of feed raw materials are calculated through the FISH-e tool (FISH-e), which can be used to calculate the global GHG emissions from aquaculture and mariculture.

2.2.2. GHG emissions for on-farm energy use

On farms, various energy sources including electricity and fuels are used for a range of purposes including powering vehicles and equipment, lighting, and pumps for water intake and drainage. According to MacLeod et al. (2019), the EI is calculated from the amount of electricity, diesel and petrol used per tonne of live weight (LW) multiplied by emission factors. The EF for electricity is the value for East Asia (0.213), and the GHG emissions for energy use were calculated using Equation (3) thus:

$$GE_{ei} = EI_{ei} \times P_i \quad (3)$$

where GE_{ei} is the GHG emissions (kg) for on-farm energy use, EI_{ei} is the emission intensity of electricity, diesel and petrol per tonne of live weight (LW), and P_i is the production (tonnes) of each species group i.

2.2.3. GHG emissions for aquatic N_2O

N_2O is one of the most important greenhouse gases contributing to global climate change with a warming potential 298 times higher than of CO_2 over a 100-year time horizon (Myhre et al., 2013). N_2O is mainly generated from microbial nitrification and denitrification (Hu et al., 2012; MacLeod et al., 2019). A range of physico-chemical factors drive rates of N_2O generation including temperature, salinity and pH and these can change seasonally (Sun et al., 2013; Welti et al., 2017; Yang et al., 2021), making it difficult to quantify the emissions of N_2O from aquaculture and mariculture. Despite these difficulties, MacLeod et al. (2019) calculate the N_2O EF (0.79 $kgCO_2e$/kgLW production). The amount of N_2O was converted into CO_2 (CO_2e), where CO_2e represents other GHGs converted into CO_2 according to their warming potential. So the aquatic N_2O amount (kg) is calculated by multiplying the production by the N_2O EF per kg of production, as shown in Equation (4) thus:

$$GE_{Ni} = EF_{Ni} \times P_i \tag{4}$$

where GE_{Ni} is GHG emissions for aquatic N_2O (CO_2e), EF_{Ni} is the 0.79 $kgCO_2e$/kgLW production, and P_i is the production for each species group.

2.2.4. GHG emissions for fertilizers

Due to the lack of data on fertilizer use in aquaculture and mariculture, the GHG emissions from fertilizer was calculated using Equation (5).

$$GE_{fei} = \frac{GE_{wfei}}{P_{wi}} \times P_i \tag{5}$$

where GE_{wfei} is the amount (kg) of global pond fertilizer emissions for each species group i of and P_{wi} is the global production (kg) of each species group i. This equation is calculated only for cyprinids, general freshwater fish, shrimp and prawns, and tilapia because the amount of fertilizer used on-farm for the cultivation of bivalves, catfish and marine fish is relatively small.

2.2.5. GHG emissions for China's aquaculture and mariculture industries

In short, the carbon footprint is defined as the sum of GHG emissions from each of the four phases of production as shown in Equation (6) thus:

$$GE_i = \sum_{s=1}^{s=n} GE_{fsi} + GE_{ei} + GE_{Ni} + GE_{fei} \tag{6}$$

Where n is the number of feed raw materials for species i, based on the GE_i and the production of each species group, the EI for each species group can be calculated using Equation (7) thus:

$$EI_i = \frac{GE_i}{P_i} \tag{7}$$

and the EI for total aquaculture production can be calculated using Equation (8) thus:

$$EI = \frac{GE}{P} \tag{8}$$

where GE is the GHG emissions from all species and P is the production of all species.

2.3. The relationship between regional GDP and GHG

To explore whether GHG emissions are affected by the level of economic development, the correlation between GHG and the gross domestic product (GDP) for each province (regional GDP) was calculated. The regional GDP for mainland China (excluding Hong Kong, Taiwan and Macau) was extracted from China Statistical Yearbooks. The amount of GHG emissions from each province was log transformed based on whether the raw data was normally distributed. The Pearson or Spearman correlation coefficient was calculated using R software (R version 4.1.1). Because there are only 11 provinces with mariculture on the coast of China, we divided the analysis into coastal aquaculture and mariculture, inland aquaculture, and China's total aquaculture and mariculture.

3. Results

3.1. GHG emissions and emissions intensity from Chinese aquaculture

The nine species groups in four production phases were included in our analysis, and their average annual production for 2011 to 2020 was 41 Mt (Table 1), which accounted for 86% of the total aquaculture and mariculture production. The production of cyprinids was the highest, at 48% of total production. Freshwater bivalves (F-bivalves) had the lowest percentage of the total production at just 0.24% (Table 1). The GHG emissions (112 Mt, Fig. 1) were calculated for 2011 to 2020 for each of the nine major aquaculture and mariculture species groups. Cyprinids culture accounted for the highest percentage of emissions at 47%. The emissions of some groups correlated closely with their production. However, for some industries, particularly for marine bivalves (M-bivalves) which accounted for 30% of production, GHG emissions only accounted for 13% of the total, so production was relatively higher than their GHG emissions. Conversely freshwater shrimp (F-shrimp) accounted for 5.3% of total production but contributed 13% of GHG emissions while marine shrimp (M-shrimp) accounted for 2.7% of total production but 7.5% of GHG emissions (Table 1).

Production of feed contributed 52% of the total GHG emissions and this was the highest proportion of GHG emissions of the four phases. N_2O was also an important source of GHG emissions, contributing 29% of the total (Fig. 1). Emissions from fertilizer production within the ponds accounted for only 1.1% of GHG emissions. The GHG emissions of feed mainly increased from the cyprinids, which represented 30% of the total GHG emissions (feed, N_2O, energy-use and fertilizers). While the cyprinids also represented 14% of the total emissions (feed, N_2O, energy-use and fertilizers) by N_2O emissions (Fig. 1).

GHG emissions from feed can be divided into nine component processes. Of those nine processes, crop energy use (19%); crop LUC (18%); fertilizer production (17%); crop N_2O (16%) and CH_4 from rice production (11%) were the most important and accounted for 81% of the total feed related emissions (Fig. 2). Animal by-products (products of slaughter not included in the carcass weight) accounted for approximately 1.2% of the emissions from the feed industry phase.

The EI for each of the species groups ranged from 1.2 for bivalves to 6.7 for F-shrimp and 7.50 for M-shrimp production as shown in Fig. 3. The EI for fish ranged from 2.4 up to 4.3 for tilapia cultivation. Cultivation of bivalves had the lowest EI of 1.2 because there are no emissions

Table 1
The production and emissions percentage of nine major species groups (Mt of live weight).

Species	M-fish	M-bivalves	M-shrimp	F-shrimp	Other F-fish	Cyprinids	Catfish	Tilapia	F-bivalves	Total
Production	0.78	12.33	1.11	2.16	3.01	19.51	0.40	1.65	0.10	41.05
Percentage	1.90%	30.04%	2.70%	5.26%	7.33%	47.53%	0.97%	4.02%	0.24%	
Emissions percentage	1.65%	12.71%	7.48%	12.99%	10.84%	46.82%	1.10%	6.31%	0.11%	

Note: M-fish: marine fish; M-bivalves: marine bivalves; M-shrimp: marine shrimp; F-shrimp: freshwater shrimp; Other F-fish: other freshwater fish; F-bivalves: freshwater bivalves.

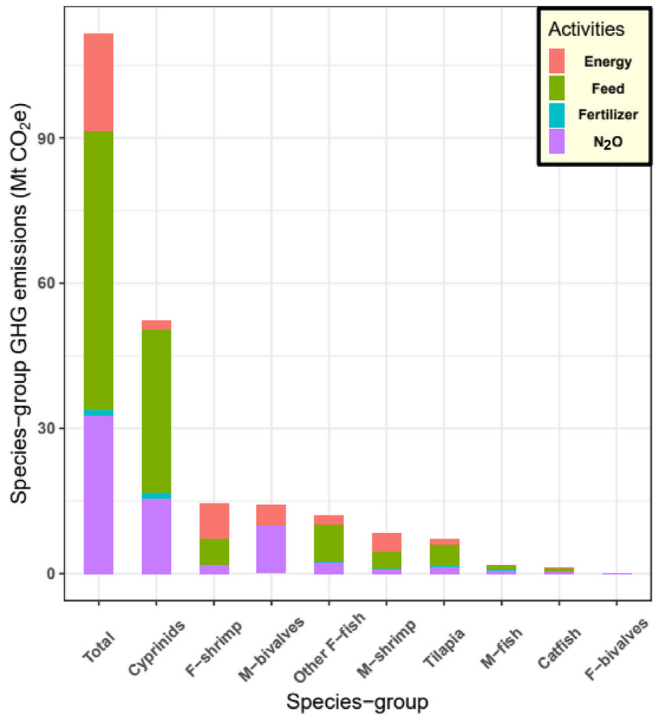


Fig. 1. The amount of greenhouse gas (GHG) emissions of nine major species groups in 2014 (M-fish: marine fish; M-bivalves: marine bivalves; M-shrimp: marine shrimp; F-shrimp: freshwater shrimp; Other F-fish: other freshwater fish; F-bivalves: freshwater bivalves).

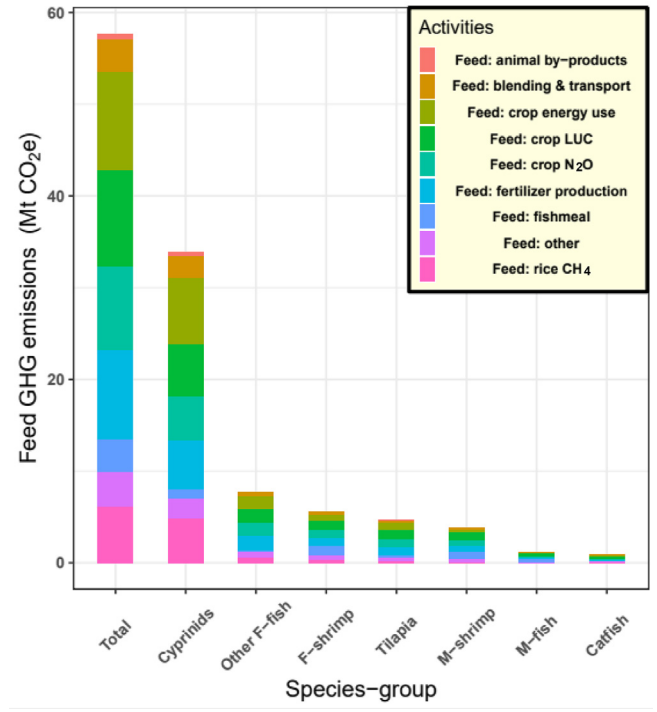


Fig. 2. Greenhouse gas (GHG) emissions of feed raw materials for the nine major species groups (M-fish: marine fish; M-bivalves: marine bivalves; M-shrimp: marine shrimp; F-shrimp: freshwater shrimp; Other F-fish: other freshwater fish; F-bivalves: freshwater bivalves). Note: M-bivalves and F-bivalves do not have GHG emissions of feed.

from feed or fertilizer. The average EI of the nine species groups was 2.7, which was similar to cyprinid cultivation. The global average EI (FAO) was 3.3, this value was between catfish and other freshwater fish, and

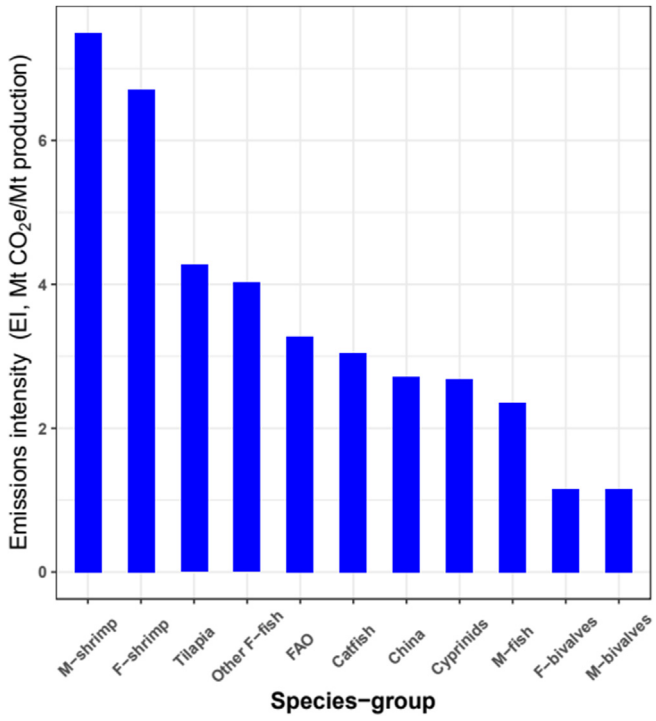


Fig. 3. Emissions intensity (EI) of nine major species groups, Chinese aquaculture (China) and global aquaculture (FAO) (M-fish: marine fish; M-bivalves: marine bivalves; M-shrimp: marine shrimp; F-shrimp: freshwater shrimp; Other F-fish: other freshwater fish; F-bivalves: freshwater bivalves).

world's EI was higher than China's EI (Fig. 3). The result showed that China will emit less GHG compare with the global average when farm the same weight of aquaculture production.

3.2. GHG emissions from 31 provinces

As shown in Fig. 4, the provinces with the highest GHG emissions are coastal or inland provinces that have more rivers and lakes, and are mostly located in eastern and southern China. The four provinces with the highest GHG emissions were Guangdong (18 Mt), Hubei (13 Mt), Jiangsu (10 Mt) and Shandong (9 Mt), and their combined emissions accounted for approximately 45% of the total from all provinces assessed in this study. The western and northern provinces have the lowest levels of GHG emissions (Fig. 4) with Qinghai, Gansu and Tibet accounting for just 0.07% of the total GHG emissions.

These results showed that emissions from feed and production of N_2O contributed the highest levels of GHG emissions, with emissions due to fertilizer use being lowest (Fig. 5). The differences in GHG emissions between provinces are related to their geographic location (whether they are close to rivers or the ocean) which determines the kinds of industries those provinces support, and their relative sizes.

3.3. Relationship between GHG emissions and regional GDP

GHG emissions had a statistically significant effect on the relative Regional GDP of the 31 provinces ($p < 0.05$), and this in turn affects the GDP. For instance, Fig. 6 shows that for China's aquaculture industry, the correlation of regional GDP was positively correlated with GHG emissions and was highly significant ($p = 0.00027$) because it was for inland provinces ($p = 0.00034$), but not for coastal provinces where it was only significant at $p < 0.05$ but not at $p < 0.01$. For the aquaculture of inland provinces (without marine aquaculture), they had the highest correlation ($R = 0.71$). Meanwhile, the GHG emissions had a greater impact on GDP of inland provinces. The 11 coastal provinces also showed a high

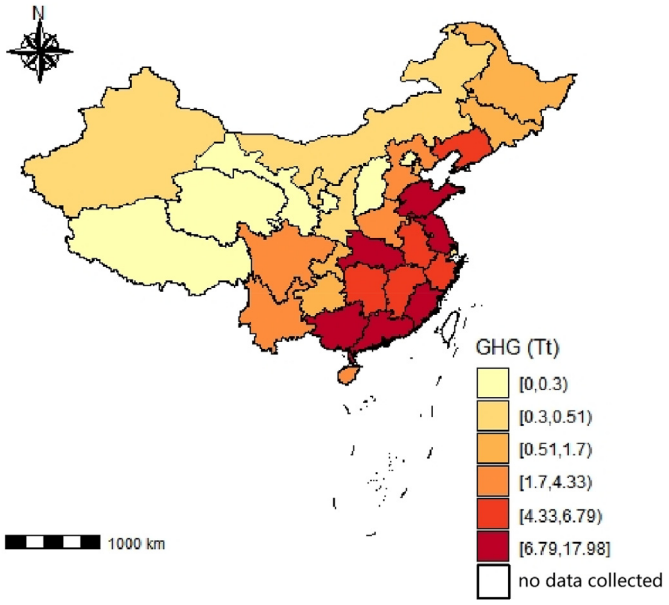


Fig. 4. Provincial spatial pattern of greenhouse gas (GHG) emissions from aquaculture in China (excluding Hong Kong, Taiwan and Macau).

correlation (R = 0.64) between regional GDP and the levels of GHG emissions. China's aquaculture (freshwater aquaculture and marine aquaculture of all provinces) had a lower correlation with regional GDP (R = 0.61) compared with independent analysis of coastal provinces (freshwater aquaculture and marine aquaculture) or inland provinces (freshwater aquaculture), but the correlation between regional GDP and GHG emissions was still high (Fig. 6).

4. Discussion

With GHG emissions continuing, global climate change is receiving increased attention from scientists and the public worldwide (Ziegler et al., 2013), and many countries have adopted policies to reduce GHG emissions. For example, China is committed to achieving a "carbon peak" by 2030 and be "carbon neutral" by 2060. This is an ambitious target that requires a coordinated effort from all industries (Liu, 2021). This study calculated the carbon footprint of the Chinese aquaculture and mariculture industries and provides an evidence base case for reducing GHG emissions from those industries. In this study, 41 Mt of aquaculture and mariculture production was included, which accounts for approximately 55% of the total global production of 75 Mt, (MacLeod et al., 2020); however, the GHG emissions from China's aquaculture and mariculture industries was 112 Mt which represents just 46% of the 245 Mt of global aquaculture and mariculture GHG emissions. Therefore, China's contribution to global GHG emissions from these industries is lower than the global average. The EI of China's aquaculture and mariculture industries was 2.7, compared to the global average of 3.3 (245 Mt CO_2e/75 Mt liveweight). Plants were not included in this analysis, however, algae are usually regarded as a carbon sink and can reduce GHG's from the atmosphere (Gui et al., 2018). The emission intensity (EI) of China's aquaculture was 2.7, and the EI of global aquaculture was 3.3 (245 Mt CO_2e/75 Mt liveweight). China's aquaculture had a smaller EI than global aquaculture. The results may contribute to the production of bivalves. In China, production of marine and freshwater bivalves accounted for 30% of total production, but globally they only account for 21%. Because bivalves emit GHG from the on-farm energy used and aquatic N_2O in this study, and bivalves do not emit GHG from feed. The feeds of bivalves mainly come from the natural environment and their food directly from the water (Gentry et al., 2017; Ray et al., 2019). Meanwhile, bivalves produce similar weight of food with smaller areas than other species (Gentry et al., 2017; Ray et al., 2019).

Of the 31 provinces assessed in this study, 20 are inland provinces without mariculture industries, which makes production of freshwater species the principal form of aquatic cultivation in China. Consequently, because they constitute the largest component of aquaculture, cypinids contribute the most to China's aquaculture production as well as its GHG emissions. Cyprinids naturally occur in wide-ranging habitats in the northern temperate zone and tropical regions and have a wide variety of spawning behaviors adapting to a variety of environments (Orban and Wu, 2008), which make them a favorable choice for fishermen to cultivation. In addition, due to economic limitations, cyprinids with low prices can meet the consumption needs of people with a lower standard of living (Crona et al., 2020). Shrimp had the highest EI, and the difference of EI between marine and freshwater shrimp was the ratio of commercial feed and non-commercial feed. The reasons for the high EI for shrimp can be attributed to on-farm energy use and feed production (Fig. 1). Compared with other species, shrimp cultivation is responsible for higher GHG emissions due to increased energy use primarily for water aeration and pumping (Cao et al., 2011; Chang et al., 2017).

According to the China Fisheries Statistical Yearbook, the production and GHG emissions were positive for 31 provinces. Most of the provinces with higher GHG emissions are coastal provinces, or inland provinces where there are more rivers and lakes such as Guangdong (coastal provinces) and Hubei (inland provinces). Because aquaculture and mariculture are one of the most important sources of protein for human consumption, these industries will play an essential role in future global food security (Ren et al., 2019). Additionally, these industries also plays an important role in the development of rural economies and the

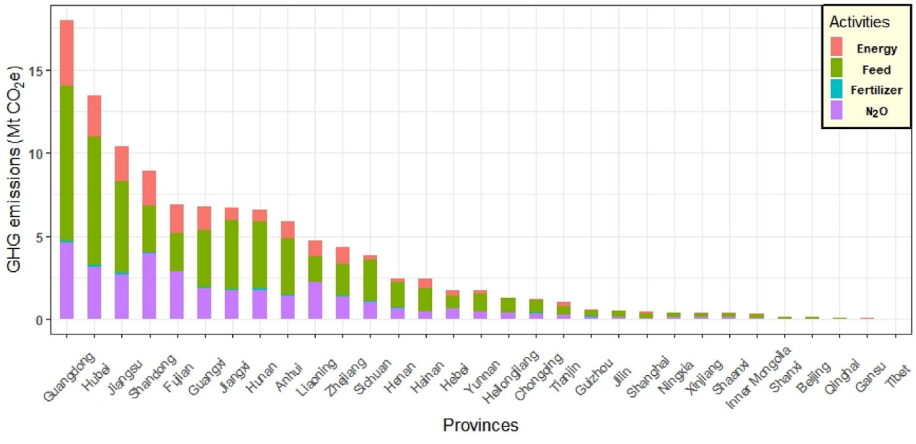


Fig. 5. Greenhouse gas (GHG) emissions and the percentage of the four activities in each of the 31 provinces.

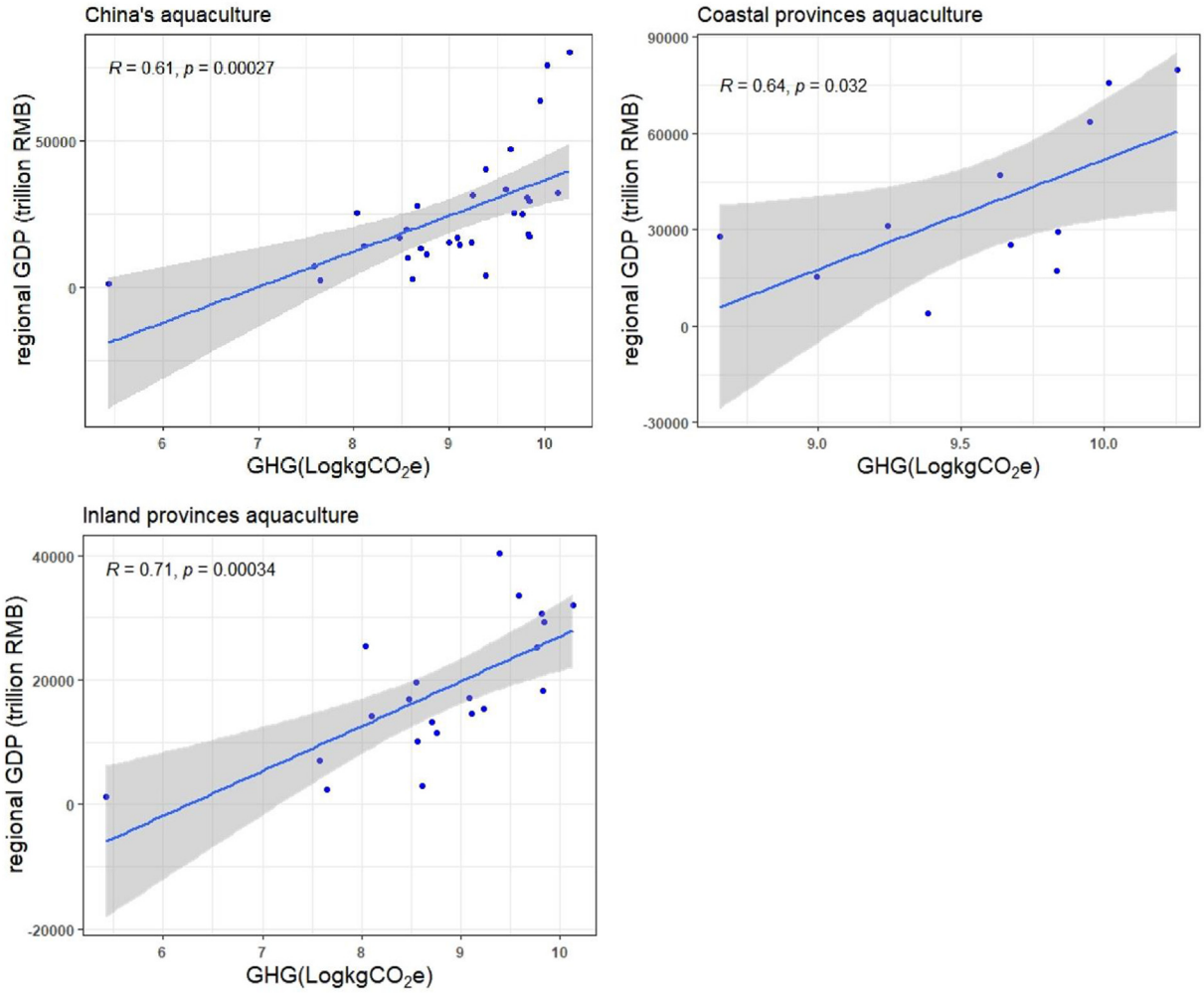


Fig. 6. The relationship between regional Gross Domestic Product (GDP) (trillion RMB) and greenhouse gas (GHG) (Log kgCO_2e) (China' s aquaculture, Coastal aquaculture, Inland aquaculture; the blue dots represent provinces; the gray area is the confidence interval).

alleviation of poverty that comes with economic development (Qi, 2002). These industries can therefore be a pathway out of poverty and into economic sustainability.

According to the findings of this study, there was a significant correlation between GHG emissions and regional GDP in all provinces. The aquaculture and mariculture industries have developed rapidly in China (Béné et al., 2016), so increased production is often accompanied by higher regional GDP. Because the relationship between aquaculture and mariculture production, and GHG emissions was positively correlated, the higher the level of production, the higher the GHG emissions were. Additionally, increased demand resulting from population increase will further promote the development of the aquaculture and mariculture industries (Béné et al., 2016). Zhang (2015) showed that consumption levels of aquatic produce is higher in the eastern provinces than in central and western provinces. Consequently, the GDP of those eastern provinces is also higher. Rapid economic and social development leads to higher levels of consumer demand for products from both aquaculture and mariculture. This in turn drives expansion of both aquaculture and mariculture in provinces with higher regional GDP. Therefore, those provinces also have higher levels of GHG emmisions (Zai, 2010; Liu et al., 2011).

There are some limitations in the study. Although this study calculated the GHG emissions for over 85% of the average annual production for 2011 through to 2020, GHG emissions will still be underestimated due to omission from the other 15%. Also, the lack of EI's for feed raw material (low value or trash fish) of marine fish production also contributed to the underestimation of GHG emissions. Finally, studies have shown that emissions will also arise from the post-farm supply chains including distribution of processed products, refrigerant leakage, and emissions from wastewater treatment plants (Iribarren et al., 2010; Ziegler et al., 2013; Bergman et al., 2020). Emissions from these sources have not been included in this study due to the lack of available information. Nevertheless, the four phases are included in the analysis from the on-farm stage, and this is able to explain most of the GHG emissions from the aquaculture and mariculture industries (Cao et al., 2011; Song et al., 2019).

Compared with terrestrial livestock production, the aquaculture and mariculture industries are relatively new; and their annual production totals are also lower, so the total GHG emissions are lower as well (Waite et al., 2014; Tacon and Metian, 2015). Being a relatively new industry, there is more potential for technological improvement including breeding and genetic selection methods (Bergman et al., 2020). Adjusting the mix of production of species groups such as increasing the amount of bivalve production due to their low EI will reduce the overall levels of GHG emissions from the aquaculture and mariculture industries. With feed being the major source of GHG emissions, improvements in the way feed is produced and used will also reduce the levels of GHG emissions. Therefore, reducing emissions from crop production would lead to a reduction in aquaculture feed emissions as well (MacLeod et al., 2015). Reductions in energy use, or changing the source of energy to low (hydroelectricity, geothermal or nuclear power) or lower (natural or biogas) emission sources (Cao et al., 2011) will also reduce GHG emissions.

However, capital costs are still a significant barrier to emissions reduction. Some mitigation measures are expensive and/or may reducing production thereby requiring some level of governmental intervention to legislate changes to farm production profiles to reduce GHG emissions.

5. Conclusions

This study is the first to systematically assess China's overall carbon footprint from aquaculture and mariculture. The results showed that the total GHG emissions of China's aquaculture were 112 Mt. Emissions from the use of feed, and N_2O production contributed the highest levels of GHG emissions. Shrimp had the highest EI and bivalves the lowest with fish between them. The differences in EI between each of these groups was significant. This study also showed the provinces that had the highest levels of GHG emissions were the coastal provinces, or those inland provinces with more rivers and lakes.

This paper also showed that GHG emissions had a statistically significant effect on the relative Regional GDP of the 31 provinces ($p < 0.05$), and this in turn affects the GDP, and all the impacts of regional GDP on GHG emissions from aquaculture and mariculture were positively correlated. Compared to the findings of MacLeod et al. (2020), China's aquaculture had a lower EI because of the higher percentage of bivalve production. To conclude, the carbon footprint can be reduced by adjusting the relative proportion of species-group production, by reducing emissions from crop production, and by changing the current source and amount of energy use.

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